

1. (Page 17, 2) 对于任意整数 $k, i^{4k}, i^{4k+1}, i^{4k+2}, i^{4k+3}$ 的值各是多少?

2. (Page 17, 4) 验证共轭复数的下列性质: 若 z_1, z_2 为两复数, 则有

$$1. k \in \mathbb{Z} \quad i^4 = 1$$

$$i^{4k} = 1 \quad i^{4k+1} = i$$

$$i^{4k+2} = i^2 = -1$$

$$i^{4k+3} = i^3 = -i$$

$$\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}; \quad \overline{z_1 z_2} = \overline{z_1} \overline{z_2};$$

$$\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}} \quad (z_2 \neq 0); \quad \overline{\overline{z}} = z;$$

$$z + \overline{z} = 2\operatorname{Re} z; \quad z - \overline{z} = 2i\operatorname{Im} z;$$

$$z = \overline{z} \text{ 当且仅当 } z \text{ 是实数}$$

$$2. \text{ 取 } z_1 = x_1 + iy_1 \quad z_2 = x_2 + iy_2$$

$$\begin{aligned} (a) \quad \overline{z_1 + z_2} &= \overline{x_1 + x_2 + i(y_1 + y_2)} \\ &= (x_1 + x_2) - i(y_1 + y_2) \\ &= x_1 - iy_1 + x_2 - iy_2 \\ &= \overline{z_1} + \overline{z_2} \end{aligned}$$

$$\begin{aligned} (b) \quad \overline{z_1 z_2} &= \overline{(x_1 + iy_1)(x_2 + iy_2)} \\ &= \overline{(x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)} \\ &= (x_1 x_2 - y_1 y_2) - i(x_1 y_2 + x_2 y_1) \\ &= (x_1 - iy_1)(x_2 - iy_2) \\ &= \overline{z_1} \cdot \overline{z_2} \end{aligned}$$

$$(c) \quad z_2 \neq 0 \quad \overline{z_2} \neq 0.$$

$$z_2 \cdot \overline{z_2} = |z_2|^2 \neq 0.$$

$$\therefore \frac{1}{z_2} \text{ 有意义}$$

$$\text{由 (b) 知 } \overline{z_2} \cdot \frac{1}{z_2} = 1$$

$$\therefore \frac{1}{z_2} = \frac{\overline{1}}{\overline{z_2}}$$

$$\overline{\frac{1}{z_2}} = \overline{z_2} \cdot \frac{1}{z_2} = \frac{\overline{z_2}}{z_2}$$

$$1d) z = x + iy$$

$$\bar{z} = x - iy$$

$$\overline{\bar{z}} = x + iy = z$$

$$1e) z + \bar{z} = x + iy + (x - iy)$$

$$= 2x = 2\operatorname{Re} z$$

$$z - \bar{z} = x + iy - (x - iy)$$

$$= 2iy = 2i\operatorname{Im} z$$

$$1f) \text{ 必要性. } z = \bar{z}$$

$$x + iy = x - iy \Rightarrow y = 0$$

$$z = x \in \mathbb{R}$$

$$\text{充分性. } z = x \in \mathbb{R}$$

$$\bar{z} = x = z$$

$$\therefore z = \bar{z} \Leftrightarrow z \in \mathbb{R}$$

3. (Page 18, 1) 用 $a + ib$ 的形式表示下列复数

$$(1) \frac{1+i}{1-i}; (2) (1+i)^3$$

$$3. \frac{1+i}{1-i} = \frac{(1+i)^2}{(1-i)(1+i)} = \frac{2i}{1-i^2} = i$$

$$\begin{aligned} (1+i)^3 &= 1 + 3i + 3i^2 + i^3 \\ &= -2 + 2i \end{aligned}$$

4. (Page 18, 2) 求出下列复数的实部与虚部(其中 $z = x + iy$)

(1) $\frac{1}{z}$; (2) $\frac{z+1}{3z+2}$; (3) z^3

4. $z = x + iy$

(1) $w = \frac{1}{z} = \frac{1}{x+iy}$

$$= \frac{x-iy}{(x+iy)(x-iy)}$$

$$= \frac{x-iy}{x^2+y^2}$$

$$\operatorname{Re} w = \frac{x}{x^2+y^2} \quad \operatorname{Im} w = \frac{-y}{x^2+y^2}$$

(2) $w = \frac{z+1}{3z+2} = \frac{(z+1)\overline{3z+2}}{|3z+2|^2}$

$$= \frac{(x+1+iy)(3x+2-3iy)}{(3x+2)^2+9y^2}$$

$$= \frac{(x+1)(3x+2)+3y^2-iy}{(3x+2)^2+9y^2}$$

$$\operatorname{Re} w = \frac{(x+1)(3x+2)+3y^2}{(3x+2)^2+9y^2} \quad \operatorname{Im} w = \frac{-y}{(3x+2)^2+9y^2}$$

(3) $w = z^3$

$$= (x+iy)^3$$

$$= x^3 + 3x^2 \cdot iy + 3x \cdot (iy)^2 + i^3 y^3$$

$$= x^3 + 3x^2 y i - 3xy^2 - iy^3$$

$$= (x^3 - 3xy^2) + i(3x^2 y - y^3)$$

$$\operatorname{Re} w = x^3 - 3xy^2 \quad \operatorname{Im} w = 3x^2 y - y^3$$

5. (Page 18, 4) 设 $z_1 = \frac{1+i}{\sqrt{2}}$, $z_2 = \sqrt{3} - i$, 试用指数形式表示复数 $z_1 z_2$ 与 $\frac{z_1}{z_2}$.

$$5. \arg z_1 = \frac{\pi}{4} \quad \arg z_2 = -\frac{\pi}{6}$$

$$z_1 = e^{i\frac{\pi}{4}} \quad z_2 = 2e^{i(-\frac{\pi}{6})}$$

$$z_1 z_2 = 2e^{i\frac{\pi}{2}}$$

$$\frac{z_1}{z_2} = \frac{1}{2}e^{i\frac{5\pi}{12}}$$

6. (Page 18, 5) 求下列各式的值

$$(2) (\sqrt{3} - i)^5; (3) \sqrt{1+i}.$$

$$6^{(2)} (\sqrt{3} - i) = 2e^{i(-\frac{\pi}{6})}$$

$$(\sqrt{3} - i)^5 = 2^5 e^{i(-\frac{5\pi}{6})}$$

$$= 32e^{i(-\frac{5\pi}{6})}$$

$$= 32\cos(-\frac{5\pi}{6}) + i32\sin(-\frac{5\pi}{6})$$

$$= -16\sqrt{3} - 16i$$

$$1^{(3)} 1+i = \sqrt{2}e^{i\frac{\pi}{4}}$$

$$\sqrt{1+i} = \sqrt[4]{2}e^{i(\frac{\pi}{4}+2k\pi)/2}, k=0,1$$

$$w_1 = \sqrt[4]{2} \cdot e^{i\frac{\pi}{8}} \quad w_2 = \sqrt[4]{2} \cdot e^{i(\frac{\pi}{8}+\pi)}$$

$$= -\sqrt[4]{2}e^{i\frac{\pi}{8}}$$

$$w^2 = 1+i \quad w = a+ib$$

$$\begin{cases} a^2 - b^2 = 1 \\ 2ab = 1 \end{cases}$$

$$a^2 + b^2 = \sqrt{2} \Rightarrow a = \sqrt{\frac{\sqrt{2}+1}{2}} \quad b = \sqrt{\frac{\sqrt{2}-1}{2}}$$

$$\sqrt{1+i} = \pm \left(\sqrt{\frac{\sqrt{2}+1}{2}} + i\sqrt{\frac{\sqrt{2}-1}{2}} \right)$$

7. (Page 18, 8) 如果 $a, b \in \mathbb{C}$, 证明 $|a-b|^2 + |a+b|^2 = 2(|a|^2 + |b|^2)$, 并说明其几何意义.

对于 $w \in \mathbb{C}$ 有 $w \cdot \bar{w} = |w|^2$

$$a-b \in \mathbb{C}$$

$$|a-b|^2 = (a-b) \cdot \overline{(a-b)}$$

$$= (a-b) \cdot (\bar{a}-\bar{b})$$

$$= a \cdot \bar{a} - a \bar{b} - \bar{a} b + b \cdot \bar{b}$$

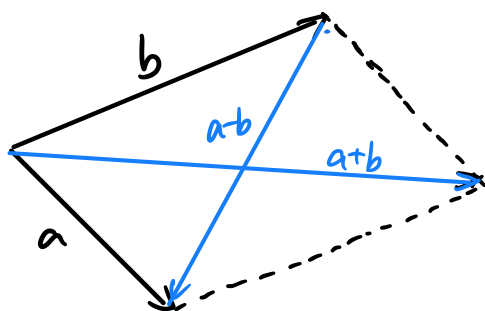
$$|a+b|^2 = (a+b) \cdot (\bar{a}+\bar{b})$$

$$= a \cdot \bar{a} + a \cdot \bar{b} + \bar{a} \cdot b + b \cdot \bar{b}$$

$$|a-b|^2 + |a+b|^2 = 2a \cdot \bar{a} + 2b \cdot \bar{b}$$

$$= 2|a|^2 + 2|b|^2$$

几何意义: 平行四边形定律



对角线长度平方和等于四条边长平方和.
对角线正好为 $a+b$ 与 $a-b$